CLASS 7 — MATHS · FORMULA SHEET

Chapter 7: **A Tale of Three Intersecting Lines** | Quick revision, one page

Triangle inequality

- Sum of any two sides $>$ the third side; difference of any two sides $<$ the third side.
- If two sides are a and b, the third side c satisfies $|a-b| < c < a+b$.

Angle sum property

- Sum of the three angles of a triangle $= 180^\circ$.
- A triangle has at most one right angle or one obtuse angle.

Exterior angle

- Exterior angle $=$ sum of the two interior opposite angles.
- Exterior angle $+$ adjacent interior angle $= 180^\circ$ (linear pair); the exterior angle is greater than either interior opposite angle.

Types of triangles

- By sides: equilateral (3 equal sides, each angle 60°); isosceles (2 equal sides, angles opposite them equal); scalene (no equal sides).
- By angles: acute (all $< 90^\circ$); right (one 90°); obtuse (one $> 90^\circ$).
- Isosceles with apex angle A: each base angle $= \dfrac{180^\circ - A}{2}$.

Perimeter

- Perimeter $= a+b+c$; for isosceles with equal sides a and base b: $P = 2a + b$.

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